NETWORK INFRASTRUCTURE DESIGN

Computer Networking Final Project Report

Presented To
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Network Design Project Report

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1. Executive Summary

This report delineates the comprehensive network topology and architectural specifications developed for Computer Networking Course Final Project. The infrastructure utilizes a robust **Three-Tier Hierarchical Model**, physically distributed between an **Enterprise Headquarters** and a remote **Branch Office**.

Architectural Stratification:

1. **Core/Edge Layer:** Four high-capacity routers form a central backbone (WAN), facilitating connectivity between the Enterprise and Branch sites, handling Network Address Translation (NAT), and enforcing centralized authentication.
2. **Distribution Layer:** Multilayer Switches (MLS) act as the aggregation points for each site, handling inter-VLAN routing and gateway redundancy via HSRP.
3. **Access Layer:** Fixed-configuration switches provide end-user connectivity, logically segmented by department (HR, Finance, Managers, IT).

Key Architectural Attributes:

- **Site Segmentation:** Logical division into **Enterprise** (VLANs 10-50) and **Branch** (VLANs 60-90) zones.
- **Security Framework:** Implementation of **AAA (TACACS+)** for access control, **SSH v2** for management, and **NAT** for securing the external Web Server.
- **High Availability:** Deployment of **HSRP** for gateway failover and **EtherChannel** for uplink aggregation.
- **Dynamic Routing:** Utilization of **OSPF Area 0** across the core backbone to ensure rapid convergence.

2. Network Topology Analysis

Topological Classification: Full Hierarchical Mesh with Multi-Site Distribution

Visual & Logical Layout:

- **Central Backbone (WAN):** A high-speed mesh of four routers (R0-R3) depicted centrally, interconnecting the two main sites.
- **Enterprise Site (Left):** Hosts the primary administrative departments and the Server Farm.
  - **VLAN 10 (HR):** Cyan Zone
  - **VLAN 20 (Finance):** Green Zone
  - **VLAN 30 (Managers):** Pink Zone
  - **VLAN 40 (IT):** Orange Zone
  - **VLAN 50 (Server Farm):** Yellow Zone (Hosting FTP, DHCP, AAA)
- **Branch Site (Right):** Mirrors the departmental structure of the Enterprise site for remote operations.
  - **VLAN 60 (HR)**
  - **VLAN 70 (Finance)**
  - **VLAN 80 (Managers)**
  - **VLAN 90 (IT)**
- **DMZ / External Services:** An HTTP/HTTPS Web Server is isolated off the Edge Router (R1) via Switch9, simulating external or DMZ connectivity.

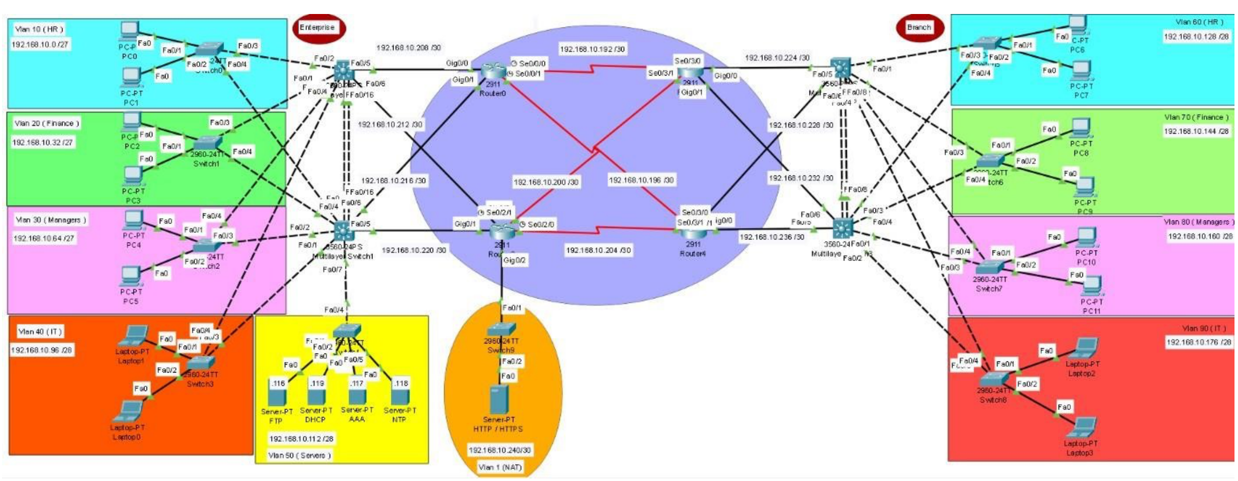

Figure 1: Complete Network Topology Diagram showing Enterprise (Left), Branch (Right), and Core Backbone.

3. Hardware Inventory

Device Designation	**Hostname**	**Model**	**Operational Role**	**Key Configured Features**
Router 0	R0	2911	Core Router	OSPF, SSH
Router 1	R1	2911	Edge/NAT Router	OSPF, **NAT (Static)**, SSH
Router 2	R2	2911	Core/Sec Router	OSPF, **AAA (TACACS+)**, SSH
Router 3	R3	2911	Core Router	OSPF, Gateway of Last Resort
MLS 0-1	MLS_0/1	3560	Enterprise Distribution	OSPF, HSRP, LACP
MLS 2-3	MLS_2/3	3560	Branch Distribution	OSPF, HSRP, PAgP
Switches 0-4	S0 - S4	2960	Enterprise Access	VLANs 10-50, Trunking
Switches 5-8	S5 - S8	2960	Branch Access	VLANs 60-90, Trunking

4. IP Addressing & Connectivity Framework

4.1. VLAN Addressing Scheme

Traffic is segmented by Department and Site.

Site	**Department**	**VLAN**	**Subnet**	**Virtual Gateway**	**Primary Forwarding Node**
Enterprise	**HR**	**10**	192.168.10.0/27	192.168.10.1	MLS_0
Enterprise	**Finance**	**20**	192.168.10.32/27	192.168.10.33	MLS_0
Enterprise	**Managers**	**30**	192.168.10.64/27	192.168.10.65	MLS_0
Enterprise	**IT**	**40**	192.168.10.96/28	192.168.10.97	MLS_0
Enterprise	**Servers**	**50**	192.168.10.112/28	192.168.10.113	MLS_1
Branch	**HR**	**60**	192.168.10.128/28	192.168.10.129	MLS_2
Branch	**Finance**	**70**	192.168.10.144/28	192.168.10.145	MLS_2
Branch	**Managers**	**80**	192.168.10.160/28	192.168.10.161	MLS_2
Branch	**IT**	**90**	192.168.10.176/28	192.168.10.177	MLS_2

4.2 Infrastructure Interconnects (Core Layer)

Point-to-Point /30 links connecting Core Routers and Distribution Switches.

Link Designation	**Interface**	**Assigned IP Address**
R0 - MLS Uplink	Gi0/0	192.168.10.209
R0 - MLS Uplink	Gi0/1	192.168.10.217
R1 - MLS Uplink	Gi0/0	192.168.10.213
R1 - MLS Uplink	Gi0/1	192.168.10.221
R2 - MLS Uplink	Gi0/0	192.168.10.225
R2 - MLS Uplink	Gi0/1	192.168.10.233
R3 - MLS Uplink	Gi0/0	192.168.10.229
R3 - MLS Uplink	Gi0/1	192.168.10.237
DMZ / Web Server	R1 Gi0/2	192.168.10.241

5. Advanced Confiiguration Details

5.1. Network Address Translation (NAT)

Configured on **Router 1 (R1)** to expose the Web Server.

- **Type:** Static NAT
- **Internal Resource:** 192.168.10.242 (HTTP/HTTPS Server)
- **External Address:** 50.50.50.2 (Public IP)

5.2. Server Infrastructure (VLAN 50)

Located in the Enterprise Server Farm (Yellow Zone):

5.2.1 FTP Server (Backup Repository):

- **Service Status:** Active.
- **Authenticated Users: Project team** (Full RWDNL permissions), cisco (Full RWDNL permissions).
- **Operational Role:** Centralized storage for network device configuration backups (e.g., R0-config, MLS_0-config).

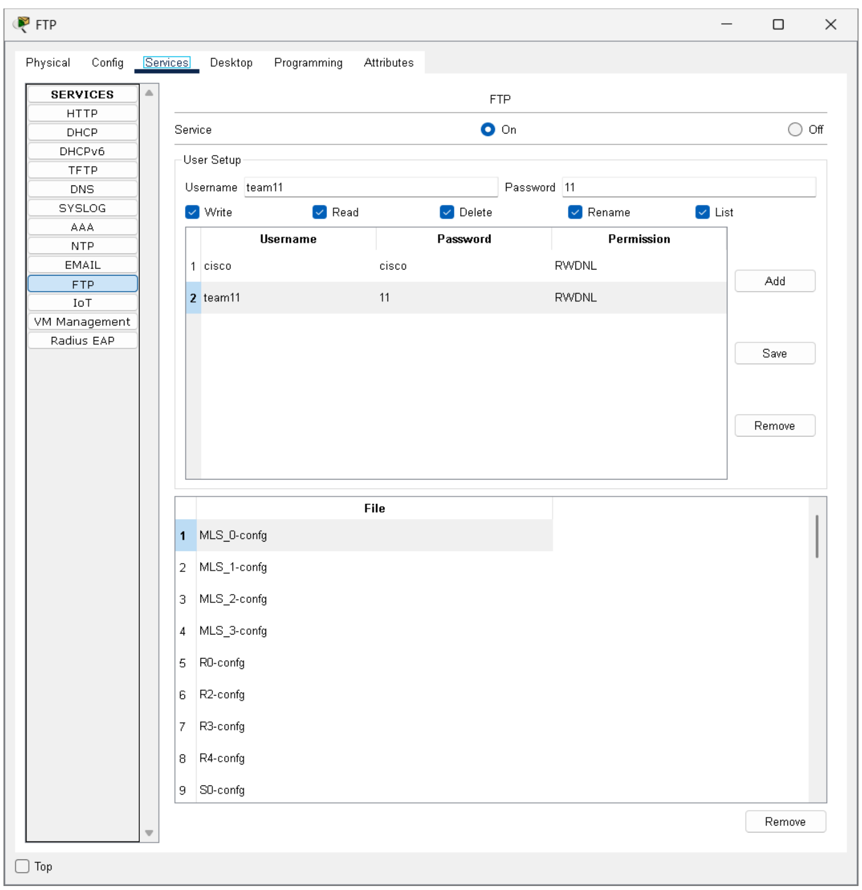

Figure 2: FTP Service Configuration showing user permissions and stored backup files.

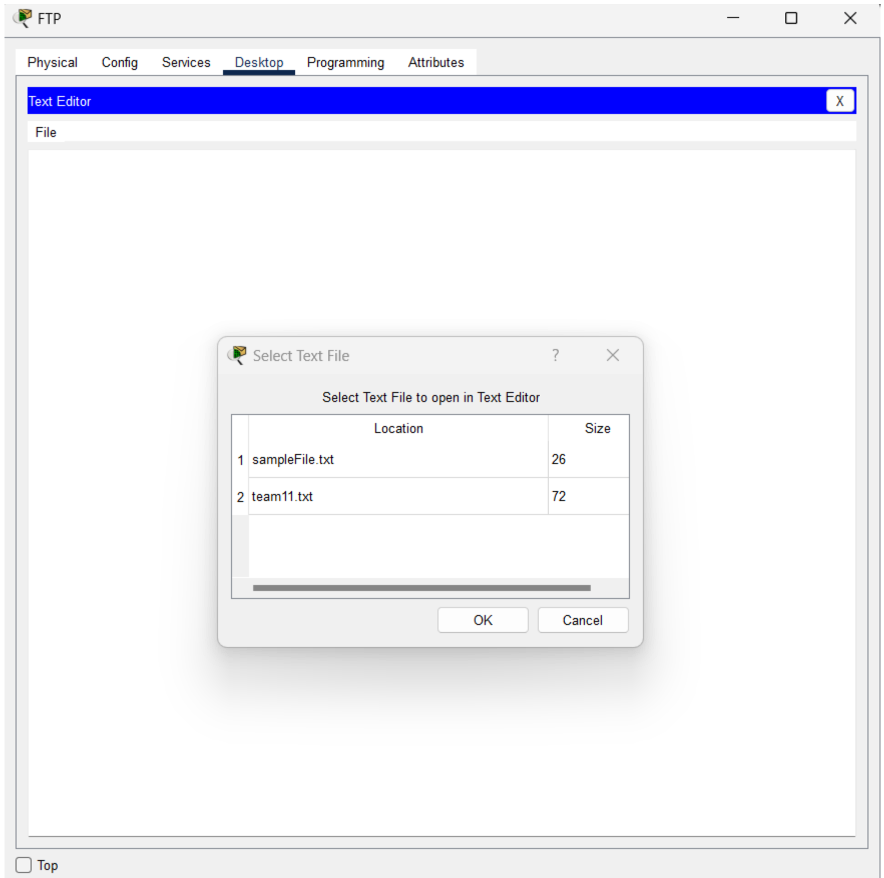

Figure 3: Creating FTP file named team.txt

Configured an FTP server to facilitate file sharing between the server and the connected PCs. On the server, we created a text file named **team.txt**

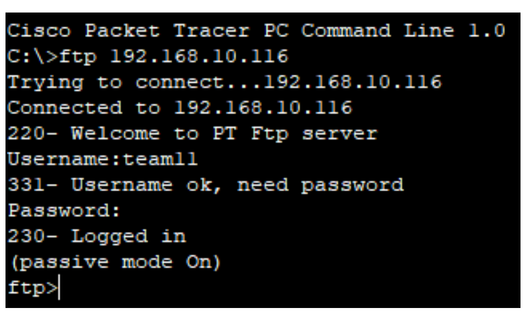

Figure 4: Pc2 retrieving team.txt file

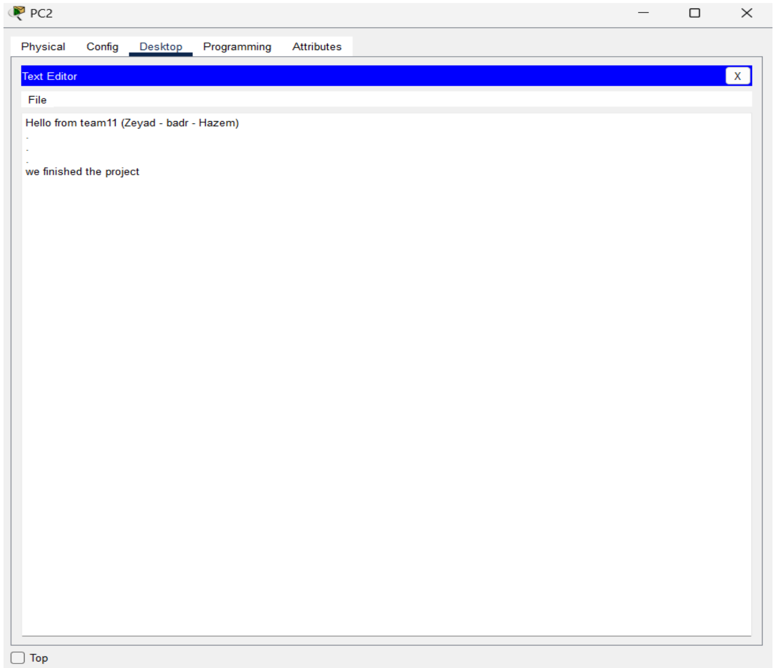

Figure 5: Open file team.txt on pc2

5.2.2 DHCP Server:

- **Role:** Centralized IP management. Requests are forwarded via ip helper-address 192.168.10.119.
- **Global DNS:** 8.8.8.8 (Configured for all pools).
- **Configured Pools:** Serves all VLANs (10-40 Enterprise, 60-90 Branch) with unique scopes.

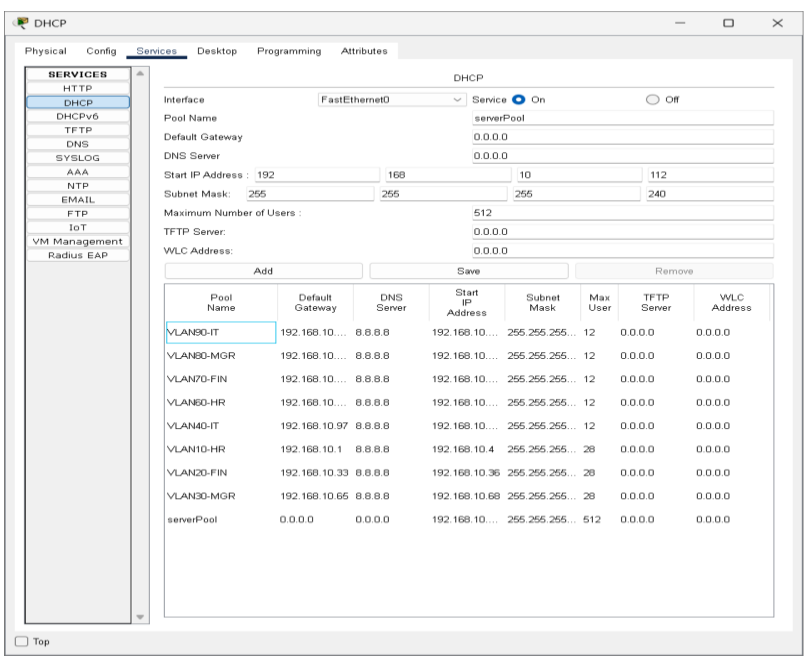

Figure 6: DHCP Pools configured for all Enterprise and Branch VLANs.

5.2.3 AAA Server (Authentication):

- **Service Type:** TACACS+.
- **Shared Secret Key:** cisco.
- **Registered Clients:**
    - Client IP 192.168.10.194 (Router 2 Serial0/3/0).
    - Client IP 192.168.10.202 (Router 2 Serial0/3/1).
- **User Database:**
    - Abdelmawla123 / Abdelmawla123
    - Abdallah123 / Abdallah123
    - hazem123 / hazem123

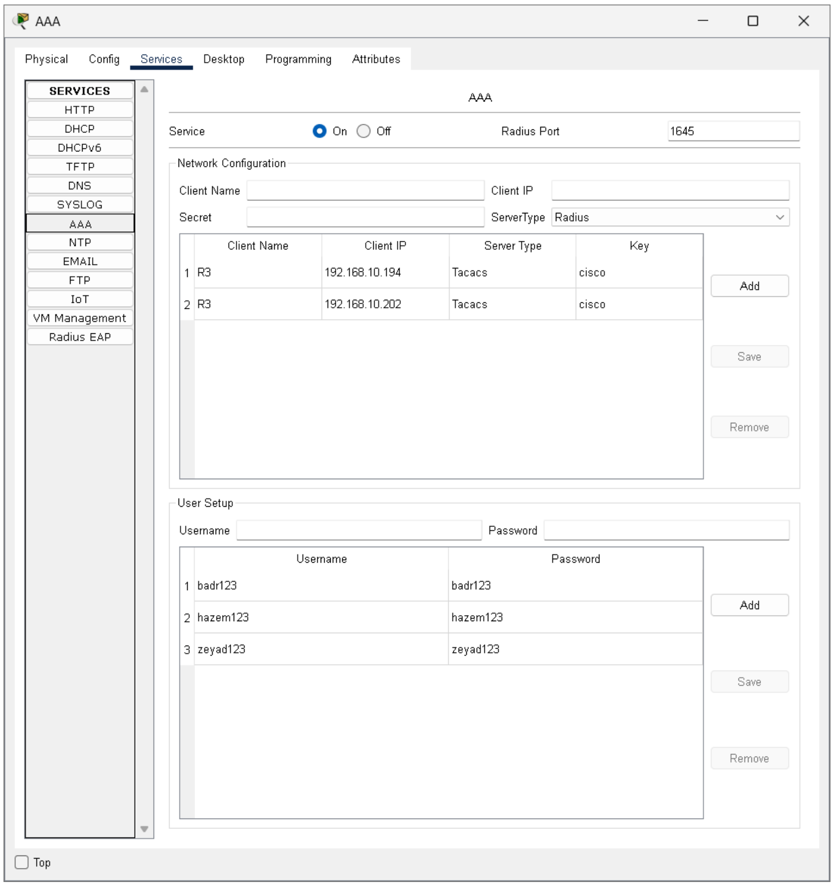

Figure 7: AAA Server configuration showing TACACS+ clients and user accounts.

5.2.4 NTP Server: the network devices (switches, core switches ,and routers) were configured to display the correct time using the Network Time Protocol (NTP).

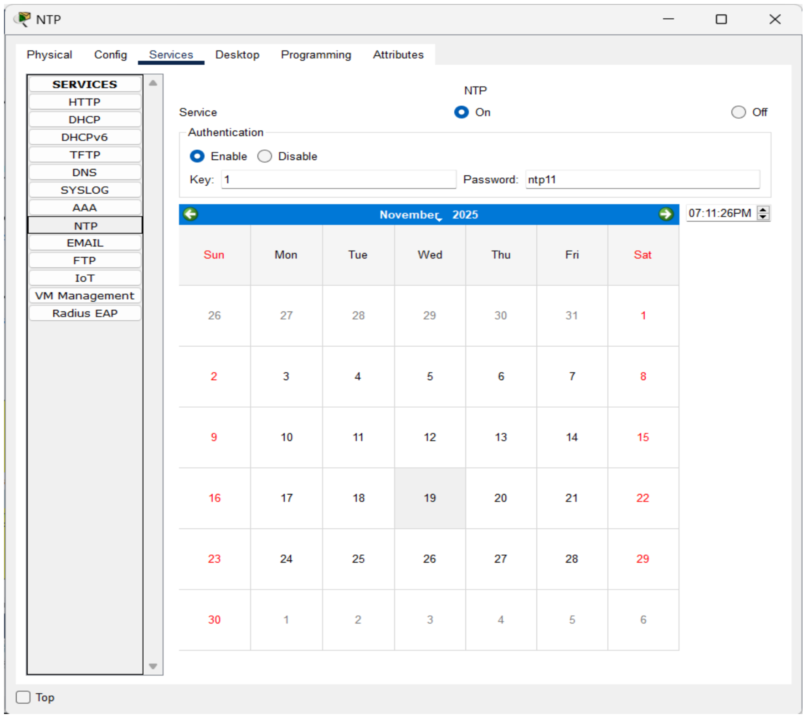

Figure 8: Creating NTP for adjusting all devices time

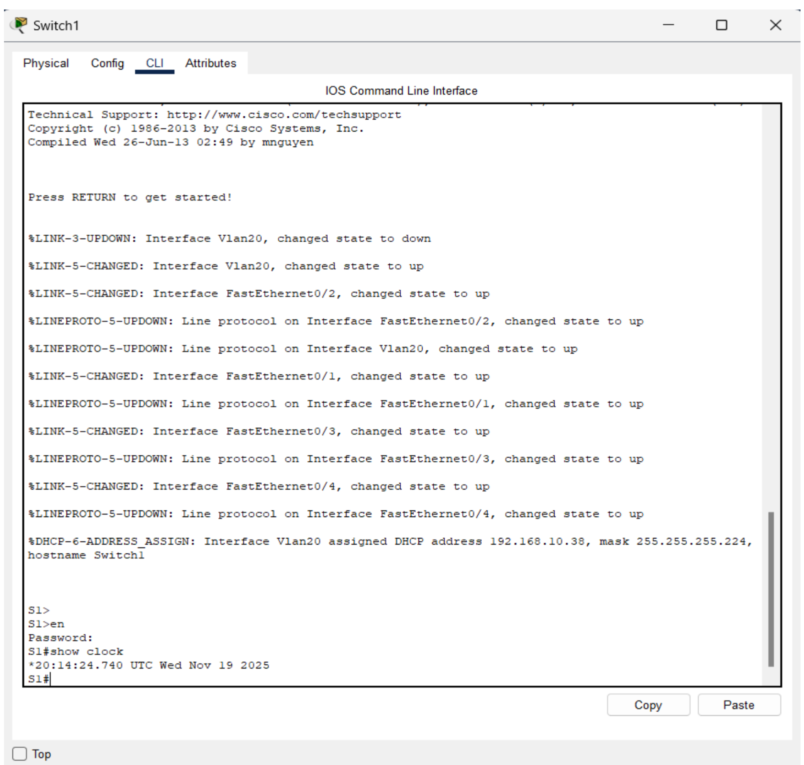

Figure 9: Time on access switch

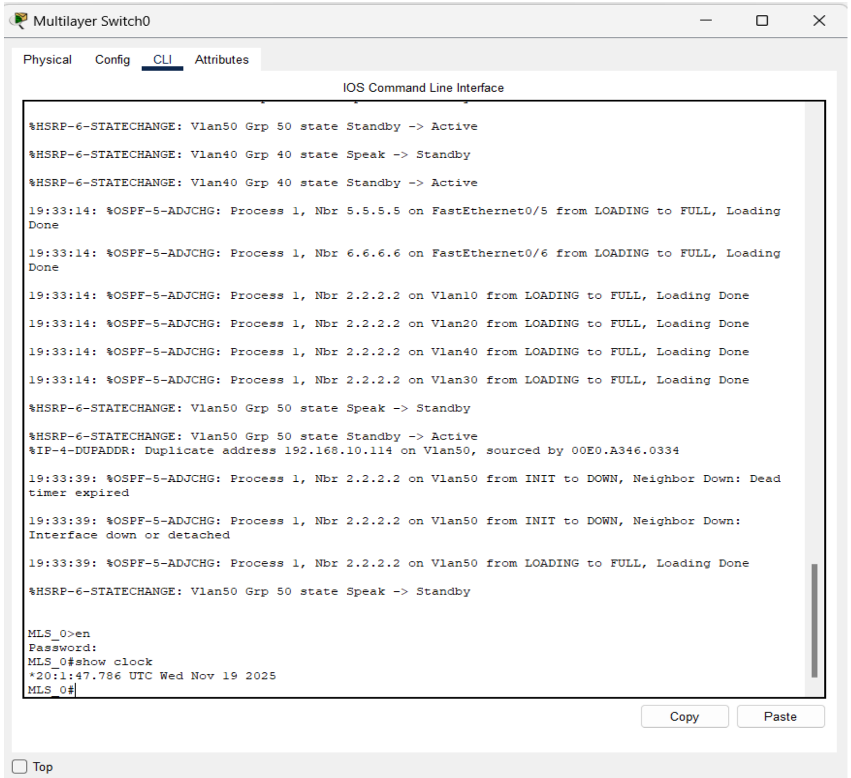

Figure 10: Time on core switch (MLS)

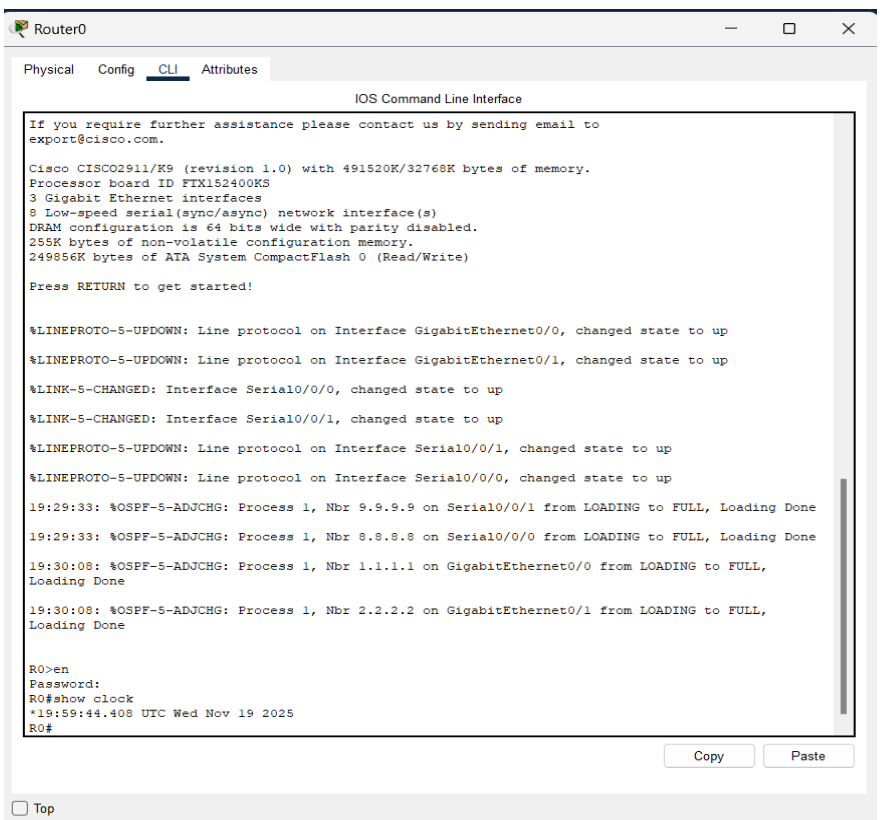

Figure 11: Time on router

5.2 Routing Strategy (OSPF Area 0)

The OSPF routing protocol acts as the unifying control plane.

- **Router Identifiers:** R0 (5.5.5.5), R1 (6.6.6.6), R2 (8.8.8.8), R3 (9.9.9.9).
- **Default Route:** Propagated by **R3** via Serial0/3/1.

5.3 High Availability

- **HSRP:** Configured on MLS pairs to ensure gateway availability for all 9 VLANs.
- **EtherChannel:**
  - **Enterprise:** LACP (Active/Passive) between MLS_0 and MLS_1.
  - **Branch:** PAgP (Desirable/Auto) between MLS_2 and MLS_3.

6 Creating enable secret password for all devices

configured the enable secret password on all network devices to secure privileged EXEC access

```
S0>en
Password:
S0#show run
Building configuration...

Current configuration : 1566 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S0
!
ip ftp username team11
ip ftp password 11
enable secret 5 $1$mERr$XZCbfAFehumrHbj7HdKg50
```

Figure 12: Enable secret password on access switch

```
MLS_0>en
Password:
MLS_0#show run
Building configuration...

Current configuration : 3191 bytes
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname MLS_0
!
!
enable secret 5 $1$mERr$XZCbfAFehumrHbj7HdKg50
```

Figure 13: Enable secret password on core switch (MLS)

```
R2>en
Password:
R2#show run
Building configuration...

Current configuration : 2072 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R2
!
!
!
enable secret 5 $1$mERr$XZCbfAFehumrHbj7HdKg50
```

Figure 14: Enable secret password on router

7 Security & Management Services

- **Remote Management:** SSH v2 enabled domain-wide (cisco.com).
- **Authentication:** Centralized **TACACS+** on Router 2.
    - **Fallback:** Local database is used if the AAA server is unreachable.
- **Administrative Accounts:** Root admin (Priv 15) and team for maintenance.

8 Connectivity Verifiication (Tests)

Evidence of successful network connectivity and service reachability.

a. End-to-End Connectivity (Ping):

Test: Ping from Enterprise HR (VLAN 10) to Branch IT (VLAN 90).

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	Laptop2	ICMP		0.000	N	0
	Successful	PC0	Laptop3	ICMP		0.000	N	1
	Successful	PC1	Laptop2	ICMP		0.000	N	2
	Successful	PC1	Laptop3	ICMP		0.000	N	3

Figure 16: Successful Ping confirming inter-VLAN and inter-site routing.

b. Web Server Access:

Test: Browser access to the external Web Server IP (50.50.50.2) from an internal PC.

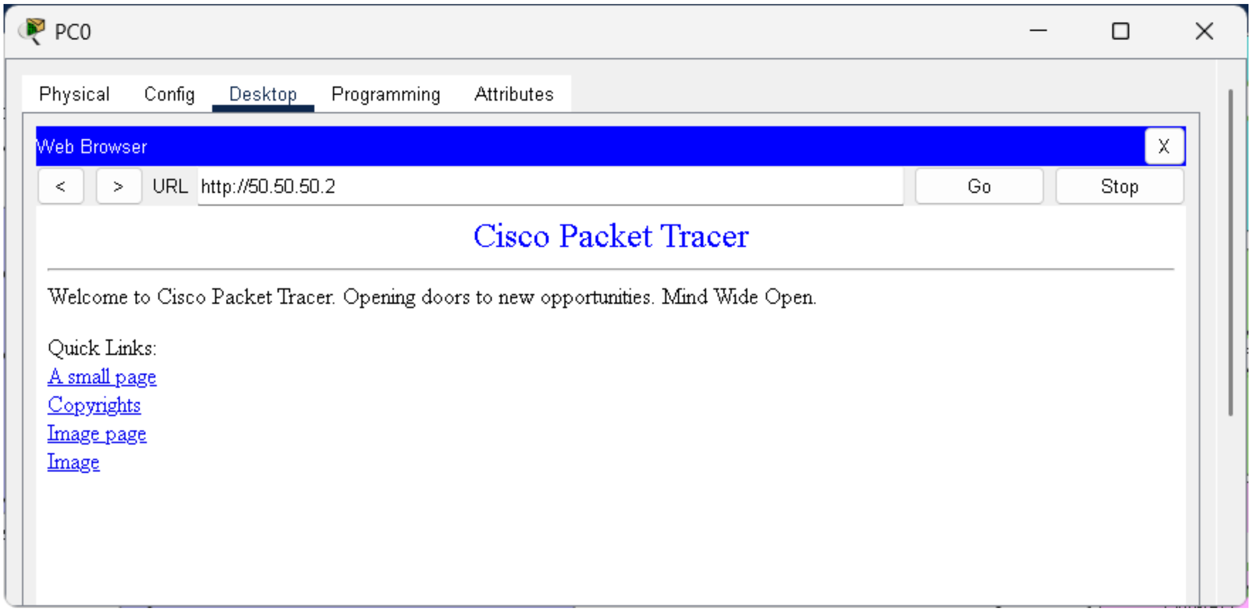

Figure 17: HTTP Access to the external web server verifying NAT and Routing.

c. Path Analysis (Trace Route):

Test: Tracer from a Branch PC to Enterprise Servers.

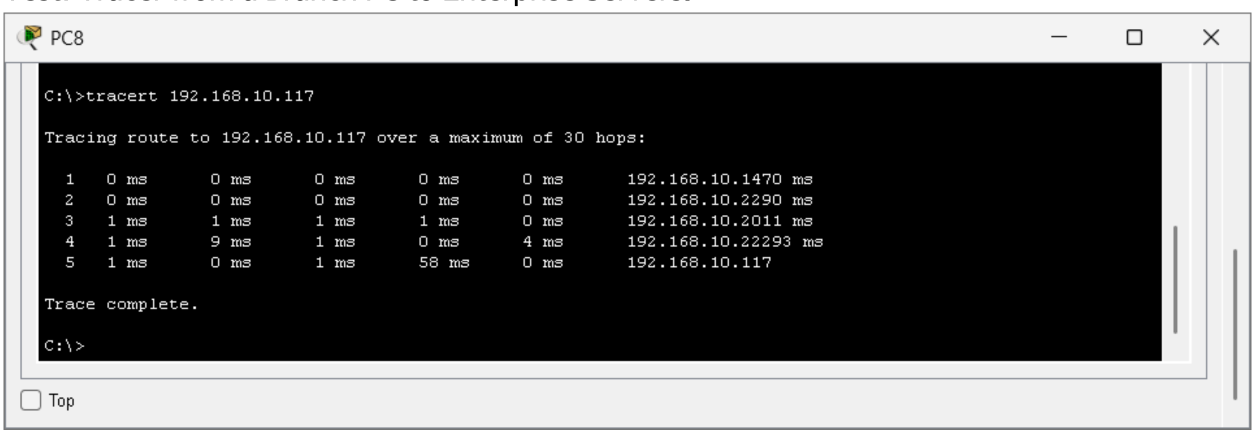

Figure 18: Trace Route showing the packet path through the Core OSPF backbone.

9 Conclusion

The network infrastructure is a highly resilient, multi-site design. By physically separating Enterprise and Branch functions and logically segmenting traffic by department (HR, Finance, Managers, IT), the architecture ensures security and performance. The inclusion of dedicated server farms, NAT for public services, and a redundant core backbone demonstrates adherence to professional network engineering standards.

10 Technical Implementation Guide (Commands Used)

This section details the specific Cisco IOS configuration commands deployed across the network infrastructure, providing an "As-Built" reference.

a. Access Layer (VLAN & Trunking)

Applicable to Switches S0 - S8.

```
interface FastEthernet0/1
 switchport mode access
 switchport access vlan [VLAN_ID]
!
interface FastEthernet0/3
 switchport mode trunk
```

i. VLAN Creation & Port Assignment:

Note: spanning-tree mode pvst and spanning-tree extend system-id are enabled by default to support Per-VLAN Spanning Tree.

ii. Management Interface (DHCP):

```
interface Vlan[VLAN_ID]
ip address dhcp
no shutdown
```

b. Distribution Layer (HSRP & EtherChannel)

Applicable to MLS_0 - MLS_3.

i. HSRP (Gateway Redundancy) Configuration:

```
    ip address 192.168.10.2 255.255.255.224
      standby version 2
      standby 10 ip 192.168.10.1      ! Virtual Gateway IP
      standby 10 priority 110         ! Higher priority for Active role
      standby 10 preempt              ! Allow takeover if priority is
higher switchport mode trunk
```

ii. DHCP Relay Agent:

```
inte interface Vlan10
 ip helper-address 192.168.10.119
```

Note: Priority is lowered (default 100) on the standby peer to ensure the correct election.

iii. EtherChannel Configuration:

```
! LACP Configuration (Enterprise Side)
interface FastEthernet0/15
  channel-group 1 mode active     ! Active LACP negotiation
!
! PAgP Configuration (Branch Side)
interface FastEthernet0/7
channel-group 1 mode desirable ! Desirable PAgP negotiation
```

c. Core Routing (OSPF)

Applicable to Routers R0-R3 and Multilayer Switches.

OSPF Process Configuration:

```
      router ospf 1
       router-id [X.X.X.X]             ! Unique ID (e.g., 5.5.5.5 for
R0)
       log-adjacency-changes
       network 192.168.10.0 0.0.0.31 area 0
       network 192.168.10.192 0.0.0.3 area 0
```

Note: All networks are advertised into Area 0 (Backbone) using wildcard masks.

d. Edge Services (NAT & Security)

i. Static NAT (Router 1):

```
! Interface definitions
interface GigabitEthernet0/0
ip nat outside
!
interface GigabitEthernet0/2
 ip nat inside
!
! NAT Translation Rule
ip nat inside source static 192.168.10.242 50.50.50.2
```

ii. AAA & TACACS+ (Router 2):

```
aaa new-model
aaa authentication login default group tacacs+ local
!
tacacs-server host 192.168.10.117
tacacs-server key cisco
```

iii. SSH Management (All Devices):

```
crypto key generate rsa
!
line vty 0 4
transport input ssh
login local
```

11 Operational Details (System Information)

This section summarizes the hardware state and licensing details extracted from the device show commands.

a. Sofitware & Licensing

- **Routers (2911):** Running IOS Version **15.1**.
  - License UDI: CISCO2911/K9 (SN: FTX1524...).
  - Features: Security/Data (supporting SSH and NAT).
- **Multilayer Switches (3560):** Running IOS Version **12.2(37)SE1**.
  - Supports ip routing, ip services.
- **Access Switches (2960):** Running IOS Version **15.0**.

b. Spanning Tree Protocol (STP)

- **Mode:** PVST+ (Per-VLAN Spanning Tree Plus) is active on all switches.
- **Root Bridge:** The Multilayer Switches (MLS) are implicitly acting as root bridges for their respective VLANs due to network topology design, though explicit root priority commands were not observed in the show run excerpts.

c. Interfiace Capacities

- **Routers:** 3x GigabitEthernet interfaces, 2x Serial WIC cards (4 Serial ports total).
- **Switches:** 24x FastEthernet ports, 2x GigabitEthernet uplinks.